



University of Bologna
Dipartimento di Informatica –
Scienza e Ingegneria (DISI)
Engineering Bologna Campus

Class of
**Infrastructures for
Cloud Computing and Big Data M**

Class Starting...
Basics, Objectives, and initial Models

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Academic year 2025/2026

CLASS WEB SITE

Find

- Teaching contents (lessons, and more...)
- Information & discussion exchange
- Some project topic and area proposals
- **Cases for practice** available when you are
- **Middleware tools there**, also individual practice
CORBA, OpenStack, Hadoop, SPARK, Kafka, Docker, Kubernetes, Orchestrators, ...

Via Web

- Many papers available
- Some personal deepening hints

virtuale.unibo.it

<https://virtuale.unibo.it/course/view.php?id=35258>

Mobile Middleware Research group

CLASS MAIN GOALS

The course aims at delivering a novel vision of all large **distributed systems** and at building a **deep, informal, practical, meditated understanding** and **experience** of their **operations**

 Since we are immersed into a pervasive ecosystem of **distributed IT systems: personally, socially, and as part of organizations**

We are interested in **building a view** based along the **system perspective**, i.e., **what is behind the architectures of big & large dimension**, and their **behavior and impact**, from two perspectives

- **First the user viewpoint**, but then more important
- **Secondly the perspective of implementers & designers**

The class focus on the **experience of long-term operations in system execution** rather than in static planning and configuration

- We aim at the entire **life cycle**, so focusing on the **execution time operations**

COURSE TARGETS

Distributed Systems are available everywhere

We must consider different features in any systems in normal environments:

- **Big data available to anyone**
- **Many processing localities available over remote data**
- **Many applications of large dimensions to support**

In general, both **company and personal accounts have reached homogeneity in the IT resource usage**

So, we have the same requirements and tools

- If some companies have a *conservative attitude* toward ICT resources,
all have also a **consolidated usage of *not on-premises resources***

COURSE TARGETS

Large global corporations that provide **Cloud services** (Amazon, Google, IBM,...)

Organization of **internal architecture** that provides Cloud services with needed Quality of Service

- Cloud Data Center Organization
- Interaction with other Data Centers and Cloud
- Intra and inter Cloud

In general, one **Cloud provider** has several local data centers and keeps them as a **central bone**, but must maintain also ***external available resources and extra-organization agreement*** for special dedicated situations

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CLOUD & BIG DATA SUSTAINABILITY

Clouds are increasingly more and more used, and much and much Big Data information is stored, kept and passed around

for sure it is feasible, but the problem is:

Is it sustainable? Now, in the future?

Sustainability is...

-
-
-
-

CLOUD & BIG DATA SUSTAINABILITY

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for sure it is feasible, but the problem is:

Is it sustainable? Now, in the future?

Sustainability is...

- Less energy loss now
- Less energy consumption now
- Low profile for more services and data
- Coordination for more sustainability
- More engagement for sustainability

DATA CENTERS FOR ONE CLOUD

Any cloud is based on **several data centers** that must reside in different **geographical areas** to create a **unique cloud**: DCs must be *far enough* to overcome local disaster, but also *far* to grant several replicated services



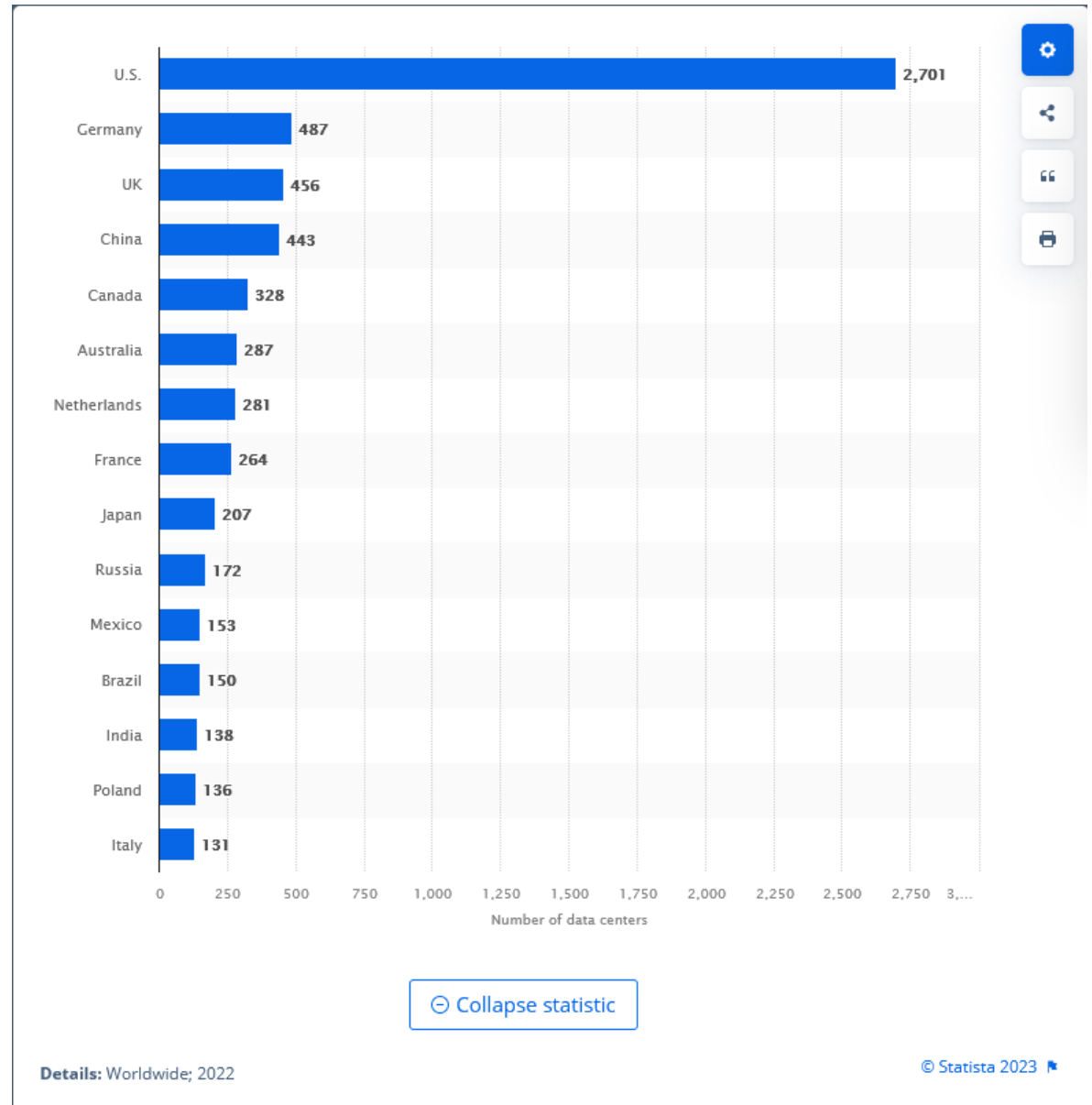
One big data center is not a cloud: a **cloud is a whole set of DCs coordinated together and cooperating for a unique quality solutions for users**

DATA CENTERS IN GOOGLE CLOUD

Many DCs are **present everywhere in the world** as in picture (2022)

Clouds consume more than **1% of electricity** in the World

They can offer their **services** as such or more **coordinated in Cloud services**



CLOUD & BIG DATA SUSTAINABILITY

Cloud Sustainability is a very Critical Issue medium – long term

- Global data centers consume around **200 TWh of electricity per year**, an amount comparable to that of **entire industrialized countries**
- The rise of **cloud, big data and AI workloads** is also increasing **water usage** for **cooling** and the overall **carbon footprint**
- The exponential growth of **digital services** makes it **necessary to redesign how infrastructures** are built and managed
- **Main challenges**
 - **Energy Consumption**
 - **Water Consumption**
 - **CO₂ Emissions and footprint**

TOWARD SUSTAINABILITY

To Make Cloud Computing More Sustainable

Renewable Energy

Cloud providers invest in **solar, wind, and long-term power agreements** to use clean energy

Low-Impact Cooling Technologies

Many **cooling solutions** to reduce energy and water consumption

Workload Optimization

Improving **software efficiency and distributing workloads** based on renewable energy availability and forecasting

Circular Economy

Recovering **waste heat from data centers** for **territorial** heating

More Transparent Metrics

New **measuring** of Power & Water User Efficiency and lifecycle **impact** help companies and governments to **make more informed decisions** (PUE and WUE)

DATA CENTER RESOURCES

Physical Architecture: Racks, Rows, and Rooms

- **Racks** Vertical cabinets containing **servers, storage arrays, network switches, and power distribution units (PDUs)**

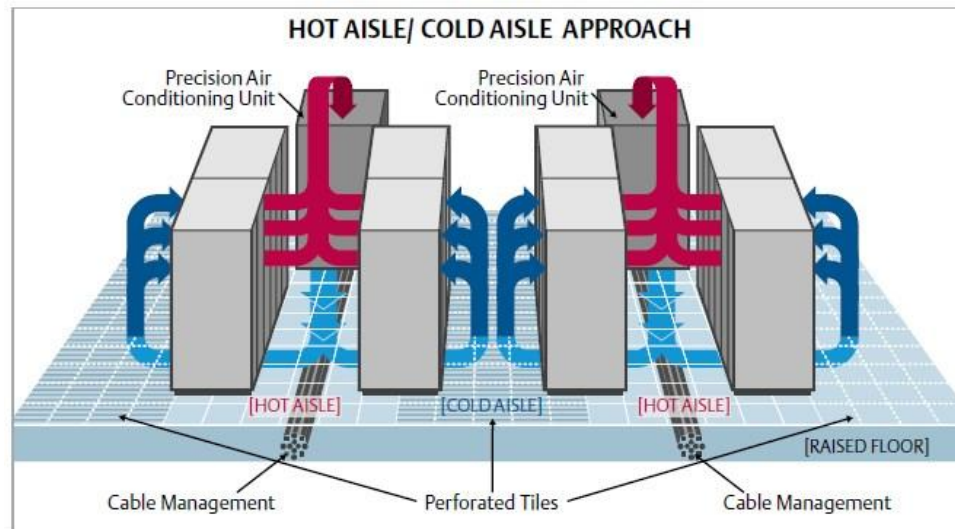
- **Rows** Racks arranged in rows

in **hot aisle / cold aisle** configured to optimize airflow and cooling

- **Rooms / Pods**

Rows grouped into **pods or cells** each with own cooling zone, power feed network aggregation layer

Physical layout affects cable length, airflow, and energy efficiency



DATA CENTER INTERCONNECTION

Traditional Three-Tier Architecture

1. Access Layer

Top-of-Rack (ToR) switches connect directly to **servers** and handle local traffic inside a rack

2. Aggregation Layer

Connects multiple racks, by providing routing, security, load balancing

3. Core Layer

High-capacity switches/routers to connects the entire data center with other data centers, the internet, WAN links

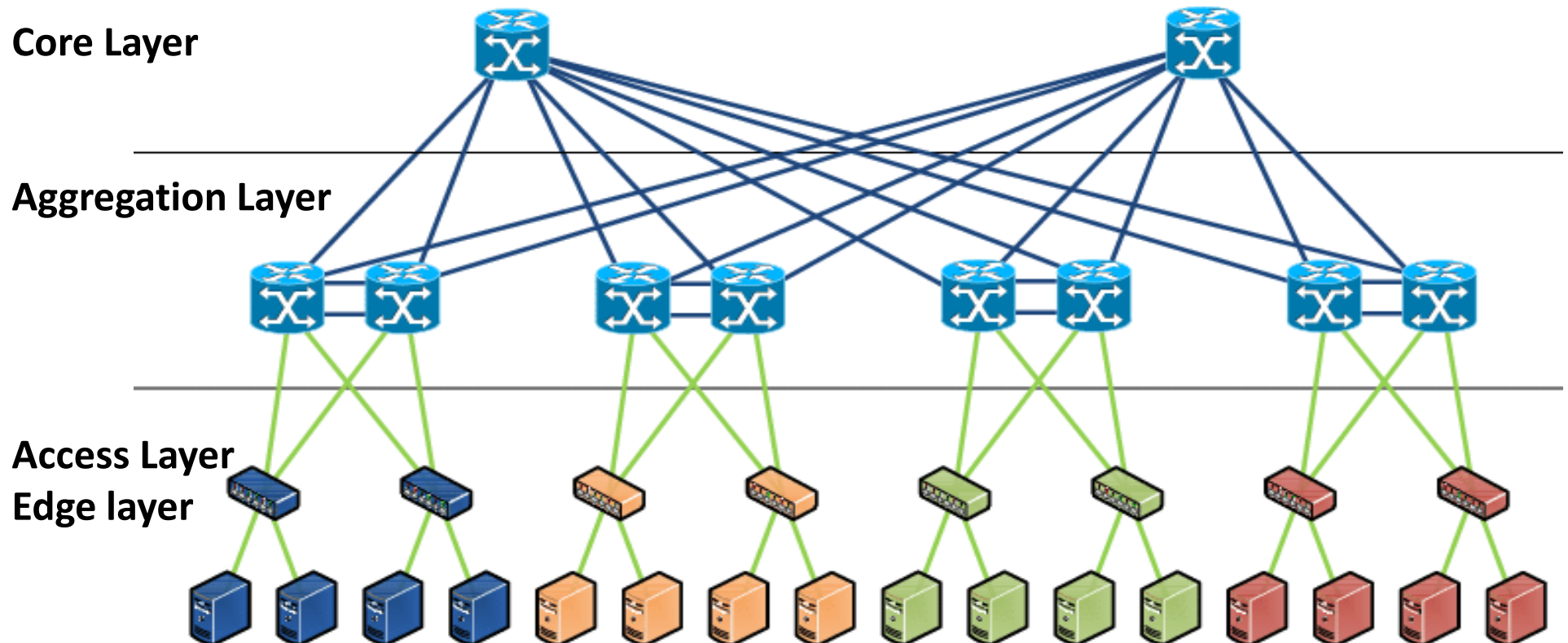
Two types of traffic Server to server (**east-west**) and external with clients, other DCs, ... (**north-south**)

Pros: Simple, well-understood

Cons: Not efficient for server-to-server traffic

3-TIER INTERCONNECTION

Very common traditional architecture



MODERN DC INTERCONNECTION

Modern Leaf-Spine Architecture

Now the dominant model for cloud and high workloads

Leaf switches

- Connect directly to **servers** and **every leaf connects to every spine**

Spine switches

- **High-bandwidth backbone** where the traffic between racks goes through

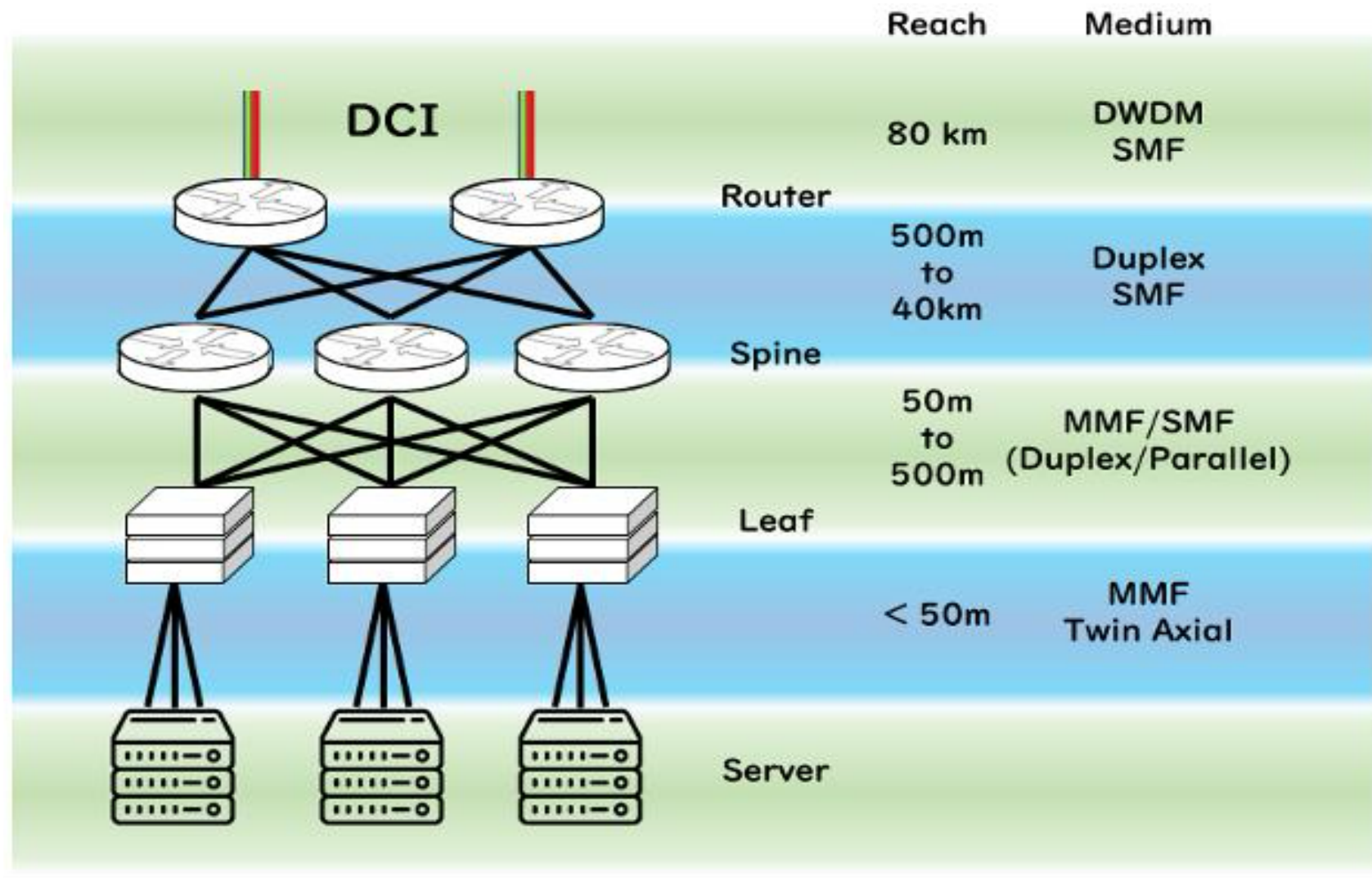
Advantages:

- Predictable latency
- High bandwidth
- Reduces switch power per bit
- Ideal for microservices and distributed systems

LEAF-SPINE INTERCONNECTION

Novel faster architecture

Fully meshed (any leaf with any spine)



COMPUTE ARCHITECTURE

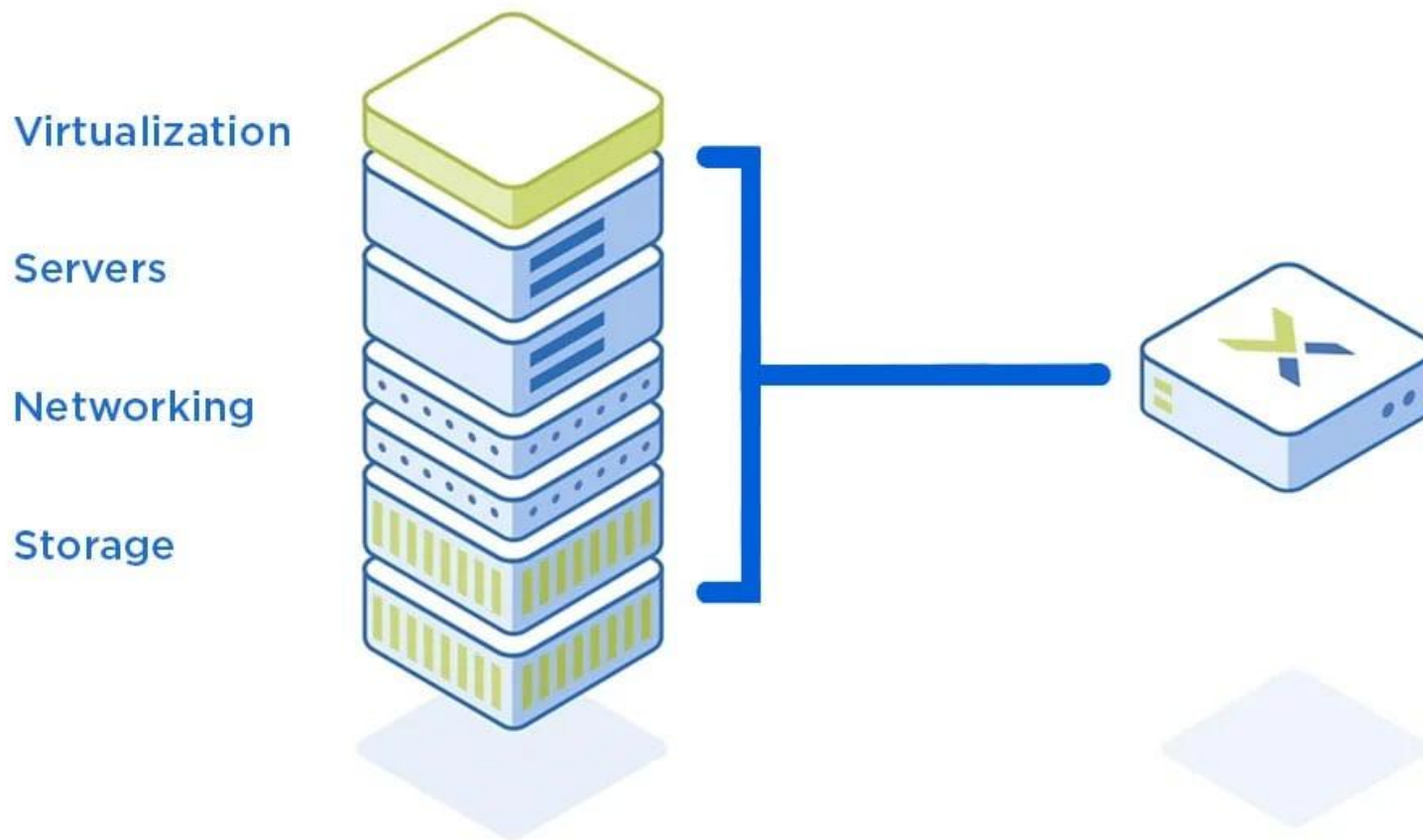
Servers are grouped in Cluster based on workload type

- **Compute clusters** **General-purpose servers** for application support to VMs, containers, microservices
- **Storage clusters** for storing data implemented by **SAN** (Storage Area Network **block-based**), **NAS** (Network Attached Storage **file-based**), **Distributed storage** (Ceph, HDFS, etc.)
- **GPU / AI clusters** **High-bandwidth interconnections** such as: InfiniBand, NVLink, RoCE (RDMA over Converged Ethernet)
- **Hyperconverged infrastructure (HCI)**
a software-defined IT framework that combines **compute, storage, networking, and virtualization** into a **single, unified system** running on commodity hardware

Different clusters have different communication patterns and energy profiles

HCI ARCHITECTURE

Hyperconverged infrastructure (HCI)



POWER MANAGEMENT

Power Architecture based on

UPS systems (Uninterruptable Power Supplies) to provide backup power either centralized or rack-level

PDU (Power Distribution Unit) to distribute power inside racks

Cooling systems Often the largest non-IT energy consumer

Many technologies: **CRAC** (Computer Room Air Conditioner) and **CRAH** (Computer Room Air Handler) **units**, **In-row** cooling, **Liquid cooling**

Internal Communication Patterns where Sustainability becomes interesting

North-South traffic

Client ↔ server goes through **core switches**

less common in modern workloads

East-West traffic

Server ↔ server **Server ↔ storage** **↔ GPU cluster**

dominates with microservices, AI, and distributed systems

IS ONE DATA CENTER ALONE SUSTAINABLE?

A Cloud is typically **based on several Data Centers working together** as a **coordinated network** instead of isolated facilities to optimize **energy use, cooling, workload distribution, and resource efficiency at a global scale**

1. Workload shifting to greener or more efficient regions

Example If wind energy peaks in Denmark at night, workloads can be shifted from a warmer region like Spain to a cooler, greener region.

Impact: This reduces carbon emissions without changing the workload itself.

2. Higher overall server utilization

When connected data centers, workloads can be consolidated into fewer active facilities

- underutilized servers can be turned off or put into deep sleep
- peak loads in one region can be balanced by spare capacity elsewhere

Impact: Higher utilization = fewer servers running = lower energy consumption

CLOUD BASED ON SEVERAL DATA CENTERS

A Cloud as a **coordinated network**

3. Cooling efficiency through geographic diversity

Cooling is one of the biggest energy drains. A **global cloud can route workloads to colder** climates, data centers with **free cooling**, facilities with **advanced liquid cooling**

Impact: Cooling energy drops dramatically.

4. Redundancy without overprovisioning

Traditional on-premise data centers must overprovision hardware for **peak demand, failover, disaster recovery**

A connected cloud spreads **redundancy** across many sites

Impact: Less hardware is needed overall, enhancing sustainability

5. Global optimization, also AI-driven

Hyperscalers use AI to coordinate predictive maintenance

Impact: The cloud becomes self-optimizing, by reducing waste at any layer

CLOUD BASED ON SEVERAL DATA CENTERS

A Cloud as a **coordinated network**

6. Better lifecycle management and circular economy

A global cloud can **reuse hardware** from one region in another one, **refurbish servers centrally**, recycle components more efficiently, deploy **modular upgrades** where they have the biggest impact

Impact: Longer hardware life and less e-waste

7. Edge + core coordination reduces unnecessary data movement

Not all workloads need to travel across the world.

A connected cloud can process **data locally at the edge** and optimize by **sending only essential results to core** data centers

Can reduce network **energy consumption**

Impact: Less data movement = lower energy use

HYPERSCALERS ARCHITECTURE

Hyperscalers (like Microsoft, Google, AWS, Meta, IBM, ...) have spent the last decade turning **data-center sustainability into engineering**

Their approach has redesigned every layer of the data-center: energy, cooling, hardware, software, and even geography

1. Energy: shifting from consumption to clean production

Hyperscalers do not just *buy* electricity but also increasingly **produce** it or lock in long-term clean energy contracts, via

- **Power Purchase Agreements (PPAs)** for wind and solar farms
- **On-site renewable generation** (solar rooftops, fuel cells)
- **Grid-interactive data centers** that shift workloads to other moments of high renewable availability
- **Carbon-free energy matching 24/7**, not just annual offsets

Why it matters

They reduce the carbon footprint of every watt consumed, and then optimize the data center itself

HYPERSCALERS ARCHITECTURE

Redesign of every layer of the data-center

2. Cooling: the biggest internal efficiency battle

Cooling is the largest non-computing energy cost enhanced by

- **Free cooling** (using outside air instead of chillers)
- **Liquid cooling** (direct-to-chip or immersion)
- **AI-driven thermal management** that adjusts cooling in real time
- **Hot/cold aisle containment** to avoid mixing air streams
- **Water-free cooling** in regions with water scarcity

Why it matters Cooling can cut total energy use by **30–40%**

3. AI-based optimization of the entire facility

Hyperscalers use **machine learning** to control server utilization, cooling setpoints, airflow, power distribution, predictive maintenance

Google famously reduced cooling energy by **up to 40%** using DeepMind reinforcement learning system.

Why it matters **Prediction** makes DC behave like a self-optimizing entity

HYPERSCALERS ARCHITECTURE

Redesign of every layer of the data-center

4. Custom hardware and software for efficiency

Hyperscalers design their own **servers** (optimized for specific workloads)

accelerators (Tensor Processing Unit or **TPUs** and other custom **AI** chips), **power supplies** (high-efficiency voltage regulators), **networking equipment**

They also use **containerized workloads**, **serverless architectures**, **autoscaling**, **workload consolidation**

Why it matters Custom hardware avoids the inefficiencies of generic servers, and smart scheduling keeps machines running closer to optimal load

5. Power distribution redesigned from scratch

Hyperscalers minimize losses by using **48V DC distribution** instead of traditional AC, by integrating **high-efficiency UPS systems**, by reducing **conversion steps between grid and server**, by placing **batteries closer to racks** (lithium-ion instead of lead-acid)

Why it matters Since any conversion wastes energy, hyperscalers remove as many as possible

HYPERSCALERS ARCHITECTURE

Redesign of every layer of the data-center

6. Geographical optimization

Hyperscalers **choose locations** based on the **whole context**: cool climates, access to renewable energy, stable grids, low water stress, proximity to fiber backbones

- Microsoft DCs in Sweden run on 100% renewable energy and use outside air cooling
- Google places facilities in regions with abundant wind energy
- AWS builds in areas with hydroelectric power

Why it matters Location alone can reduce cooling energy close to zero

7. Circular economy and lifecycle management

Hyperscalers extend sustainability by enforcing refurbishing and **reusing servers**, **recycling metals** and rare earth materials, **designing hardware for easy disassembly**, **reusing waste heat** for district heating (common in Northern Europe), using modular data-center design to reduce waste

Why it matters If the **environmental cost** of building a data center is huge, **circular design** reduces it dramatically

HYPERSCALERS ARCHITECTURE

Hyperscalers succeed because they treat **sustainability** as a **full-stack engineering requirement** and deal with any possible trajectory for optimization

Layer	Hyperscaler Solution
Energy	Renewables, Power Purchase Agreements - PAs, 24/7 carbon-free energy
Cooling	Free cooling, liquid cooling, AI-driven thermal control
Hardware	Custom servers, accelerators, efficient power supplies
Software	Autoscaling, workload consolidation, serverless
Power	High-efficiency UPS, DC distribution
Location	Climate-optimized siting, renewable-rich regions
Lifecycle	Recycling, reuse, modular construction

ANOTHER STEP ON CLOUD

Smaller data centers can work together with **larger hyperscale facilities** to create a **hybrid cloud ecosystem**

The **entire architecture connecting** both is **far more sustainable than either model alone**

Capacity of **distributing locally workload** when it is the case and going **global for optimization**

1. Local processing reduces long-distance data movement

Small, regional, or edge data centers can process fast real-time analytics, IoT data, video streams, latency-sensitive workloads

Only the essential results are sent to large hyperscale data centers

That helps sustainability

- Less long-distance network traffic
- Lower energy use in backbone networks
- Reduced load on large data centers

Effect: The whole cloud consumes less energy to move and store data

HYBRID CLOUD ECOSYSTEM

Capacity of **distributing locally workload** when it is the case and going **global for optimization**

2. Large data centers absorb heavy workloads efficiently

Hyperscale facilities are more energy-efficient, easier to be cooled and powered by more renewable energy, optimized with AI when

smaller data centers offload heavy or batch workloads to hyperscalers, the same computation is done with **much lower energy per operation**

Effect: Workloads run where they are most efficient

3. Dynamic workload balancing across the ecosystem

When **small and large data centers are connected**, the cloud can shift workloads away from overheated or overloaded small sites

- can move tasks to regions with more renewable energy
- can consolidate workloads into the most efficient facilities at any moment

This is similar how hyperscalers optimize globally, but for smaller partners

Effect: Higher utilization and lower waste across the entire network

HYBRID CLOUD ECOSYSTEM

Capacity of **distributing locally workload** when it is the case and going **global for optimization**

4. Cooling load is distributed intelligently

Small data centers often **struggle with limited cooling capacity, higher PUE, local heatwaves**

By **offloading workloads** to larger, better-cooled facilities during peak temperatures, they avoid inefficient cooling spikes

Effect: Lower cooling energy and fewer thermal emergencies

5. Shared redundancy reduces overprovisioning

A small data center alone must **overprovision hardware** for failover, peak demand, maintenance windows

But when connected to a **hyperscale cloud redundancy** is shared, backup capacity is pooled, disaster recovery is offloaded to larger sites

Effect: Less hardware, less embodied carbon, less idle energy

HYBRID CLOUD ECOSYSTEM

Capacity of **distributing locally workload** when it is the case and going **global for optimization**

6. Lifecycle and circular economy benefits

Hyperscalers have **sophisticated systems** for refurbishing servers, recycling components, reusing heat, optimizing hardware lifespan

Smaller data centers themselves **can plug into these ecosystems** instead of managing everything alone

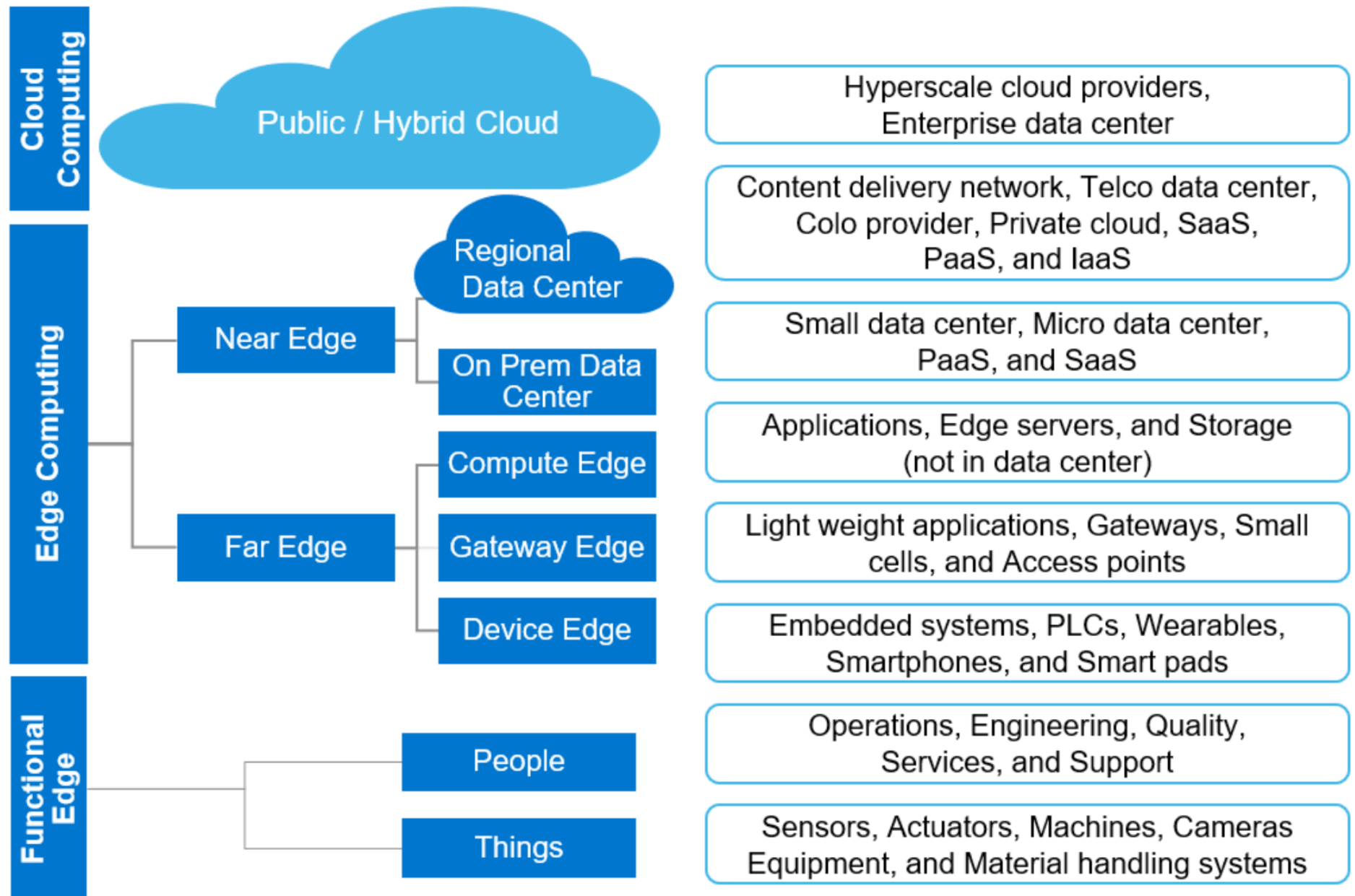
Effect: Lower e-waste and longer hardware life

7. AI-driven orchestration across mixed infrastructures

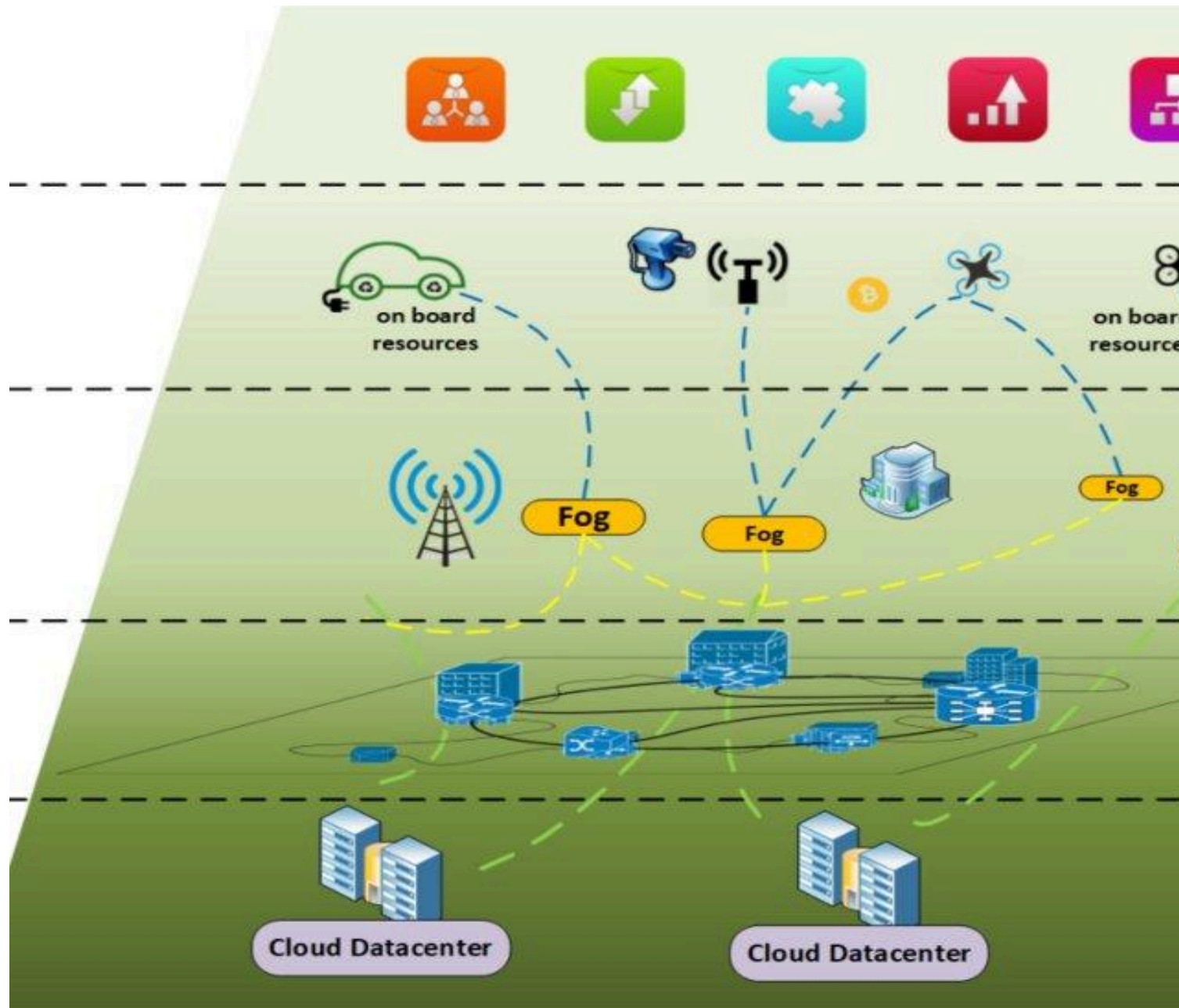
When small and large data centers are connected, AI can predict demand, route workloads optimally, adjust cooling dynamically, match workloads to renewable energy availability, detect inefficiencies across the network

Effect: The entire hybrid cloud behaves like a coordinated, self-optimizing system

LARGE CLOUD ECOSYSTEM



LARGE CLOUD ECOSYSTEM



HYBRID CLOUD ECOSYSTEM (CLOUD CONTINUUM)

Capacity of **distributing locally workload** when it is the case and going **global for optimization**

Putting it all together

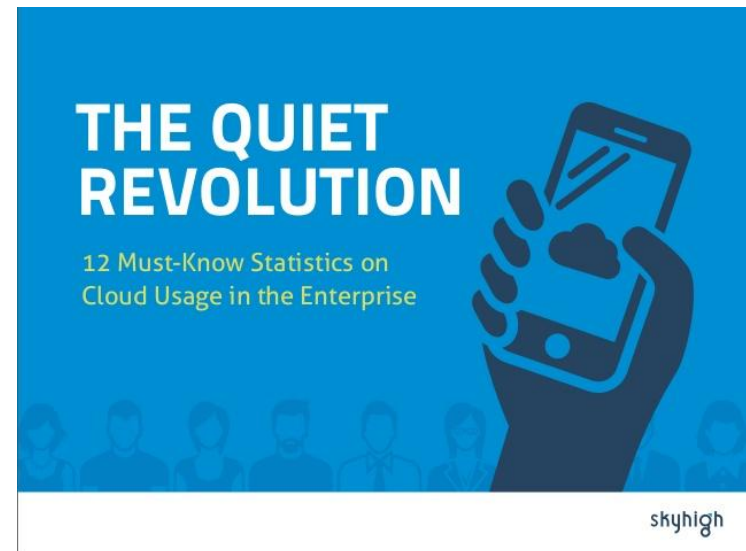
The combination of small and large data centers creates a **distributed, flexible, and efficient cloud**:

Benefit	Role of Small Data Centers	Role of Large Data Centers
Latency	Local processing	Global coordination
Efficiency	Offload heavy tasks	Run them efficiently
Cooling	Avoid peaks	Absorb load in cool regions
Redundancy	Less overprovisioning	Provide backup capacity
Sustainability	Reduce data movement	Use renewables + AI optimization

ICCBBD - CLOUD IS A REVOLUTION...

Cloud is a buzzword to be used in advertising and it is sometimes depicted as a revolution

There are many books about Cloud as a revolutionary technology



In general terms, there is no *solution of continuity* both under an organization and a technical perspective with previous experiences

Availability of remote BIG DATA Centers for services

CLOUDS ARE CHEAPER... AND WINNING...

Range in size from “edge” facilities to **megascale**

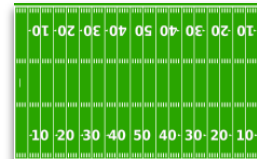
Scale economies for Data Centers

Approximate **costs** for a **small size center** (1K servers) and a **larger**, 50K server center

The bigger the better



Technology	Cost in small-sized Data Center	Cost in Large Data Center	Cloud Advantage
Network	\$95 per Mbps/month	\$13 per Mbps/month	7.1
Storage	\$2.20 per GB/month	\$0.40 per GB/month	5.7
Administration	~140 servers/Administrator	>1000 Servers/Administrator	7.1



Each **data center** is
11.5 times
the size of a football field

CLOUDS ARE LOCAL

Clouds must be available for services ‘close’ to their customers

If you are in EU, is it good to refer to very far intercontinental DATA CENTERS?

Of course, you must refer to **local resources**, not only for **technical reasons** (applied laws, market opportunity, etc.)

Many Players have developed many local data centers and they compete with those on local markets

International stakeholders, but also local ones

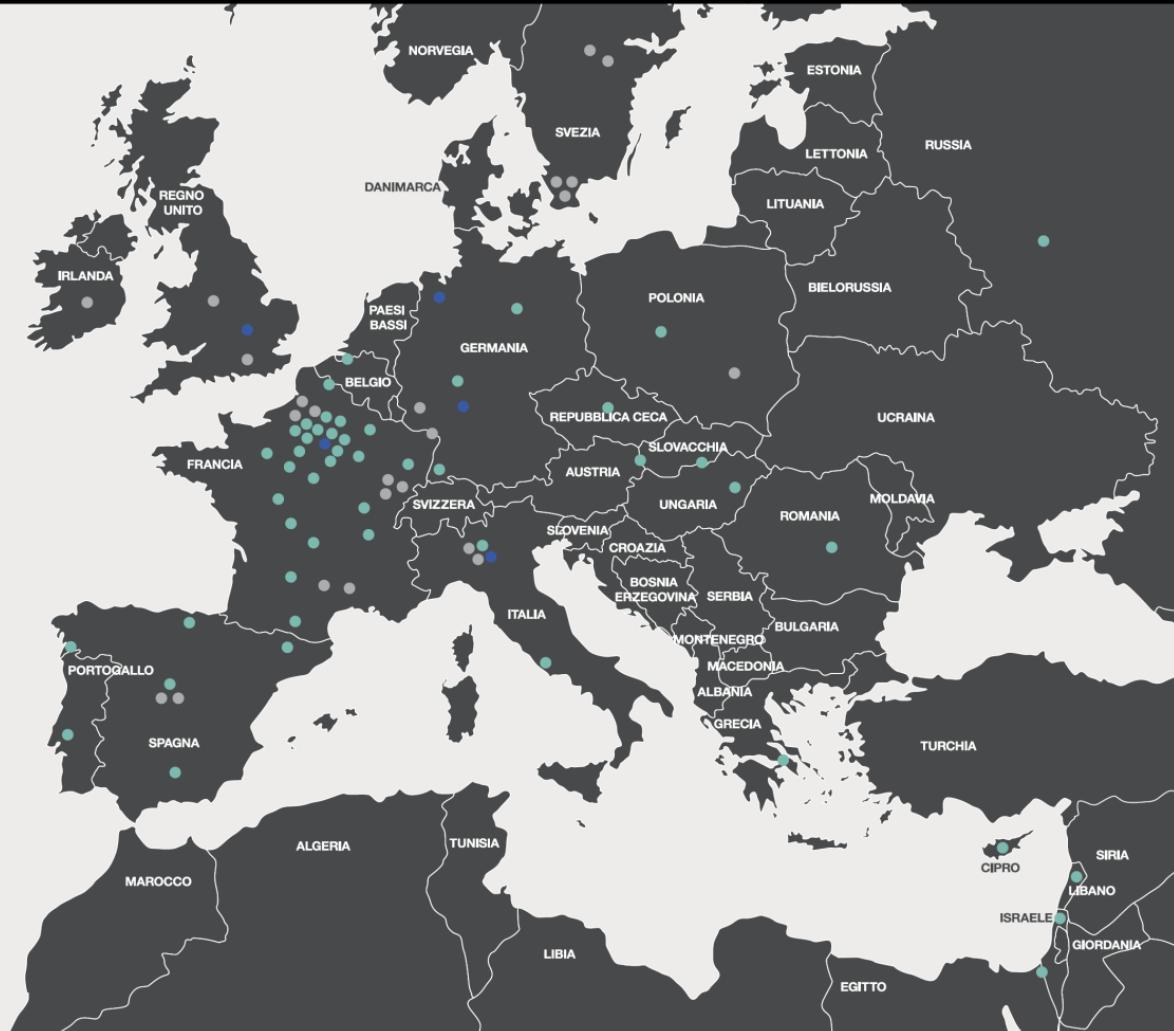
MANY LOCAL DATA CENTERS: GOOGLE DATA CENTERS



MANY LOCAL DATA CENTERS: AMAZON DATA CENTERS

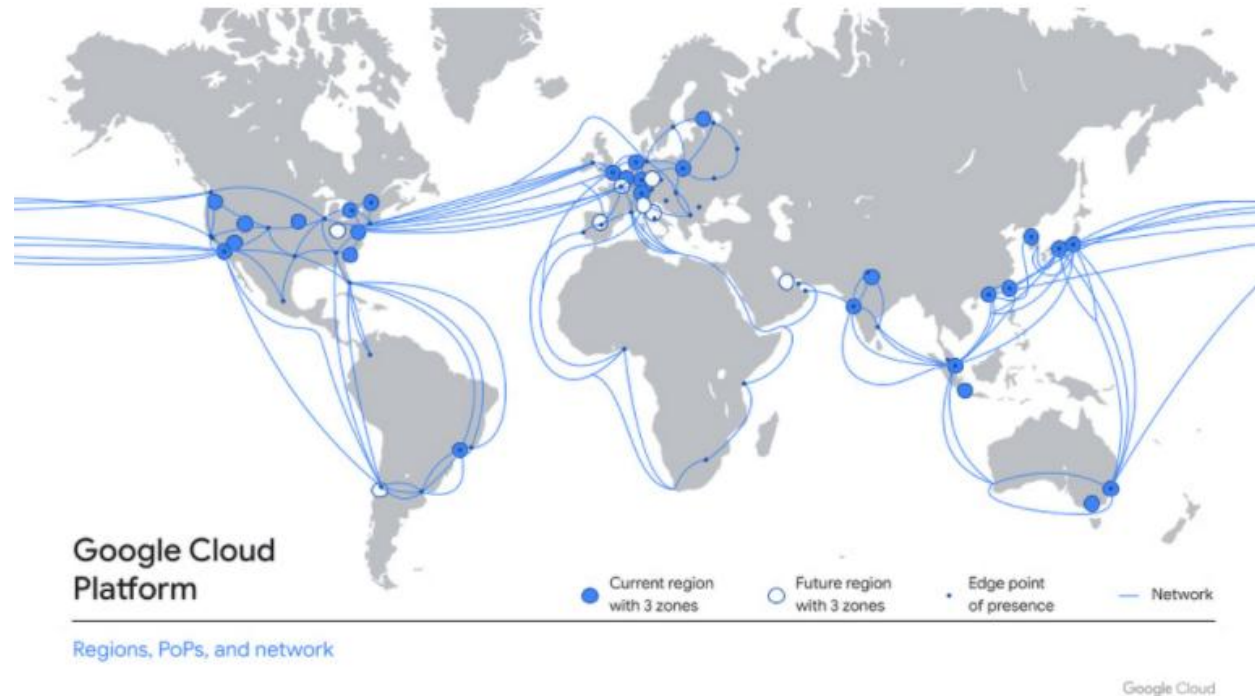


I Data Center IBM in Europa



DATA CENTERS IN GOOGLE CLOUD

The picture is for **Google** and its **growing organization** to **create the Google Cloud over many DCs**



One (even large) data center is not a cloud: a **cloud** is a **whole set of DCs coordinated and cooperating together** to provide unified services to clients

ICCB^D - BIG DATA EXPECTATIONS

DIGITALIZATION valued at growth rate of 27,6%

\$880.28 billion in 2023

to reach \$4.40 trillion by 2030

BIG DATA INVESTMENTS valued at growth rate of 14,9%

\$327.26 billion in 2023

to reach \$924.39 billion by 2030

ICT industry market estimated to be valued at

\$6.03 trillion in 2025

to reach \$7.86 trillion by 2030

driven by platform for Mobile broadband, Social business, Cloud services, and Big data and analytics

European effort

Many initiatives also within **Horizon 21-27**, also connected with

Horizon Europe, Next Generation EU, PNRR, ...

BIG DATA & MORE

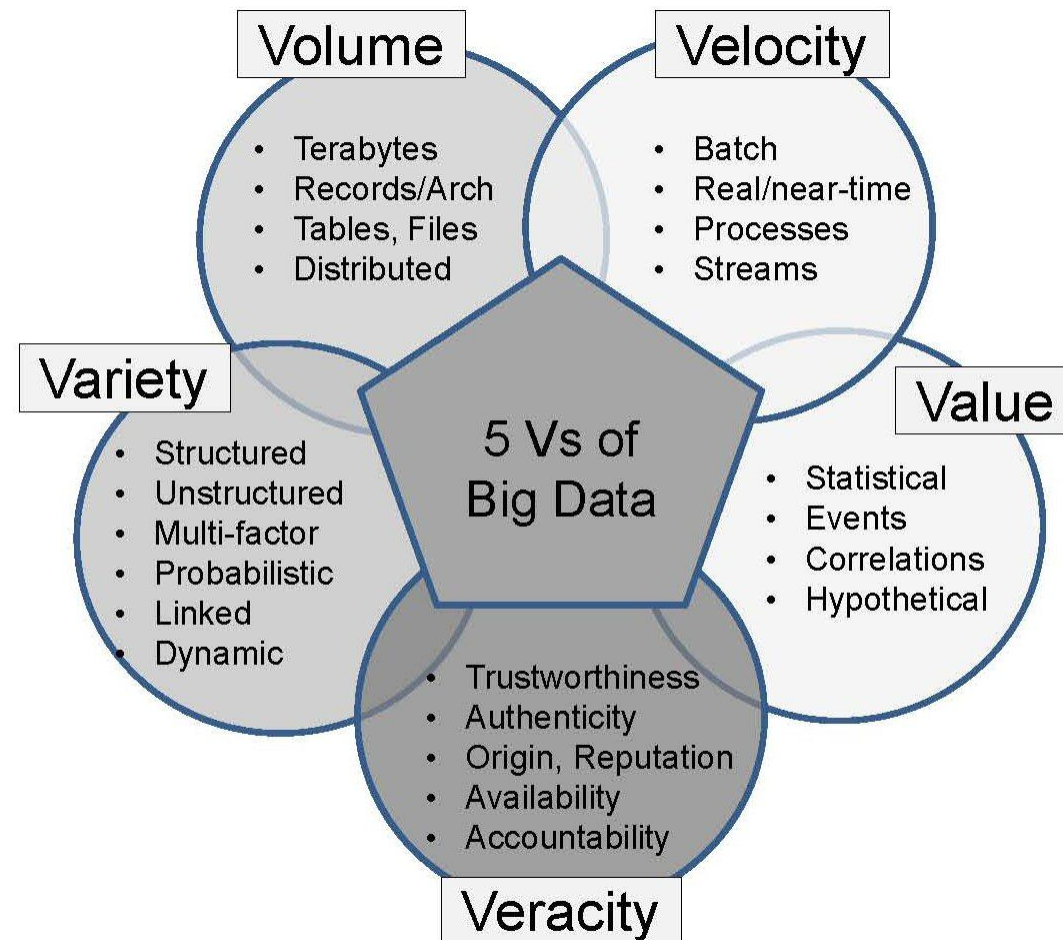
Information systems require a **quality-aware vision** that can
the organize the whole data lifecycle

5 Vs for new data processing
and
novel data treatment

- **V**olume of Data
- **V**ariety of Data
- **V**elocity
- **V**alue
- **V**eracity

6 V's also **Data Dynamicity**

- **V**ariability

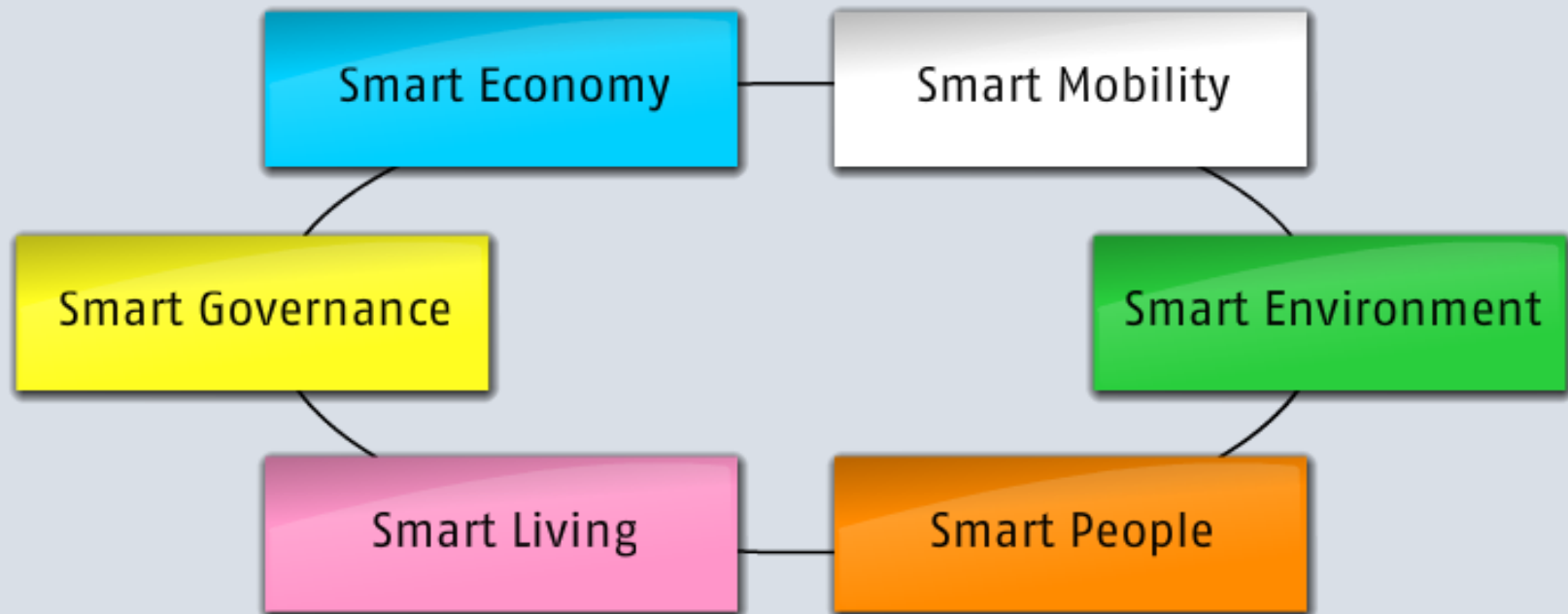


SMART CITY: SMART PERSPECTIVES

Smart cities as big data producer

The smart city model

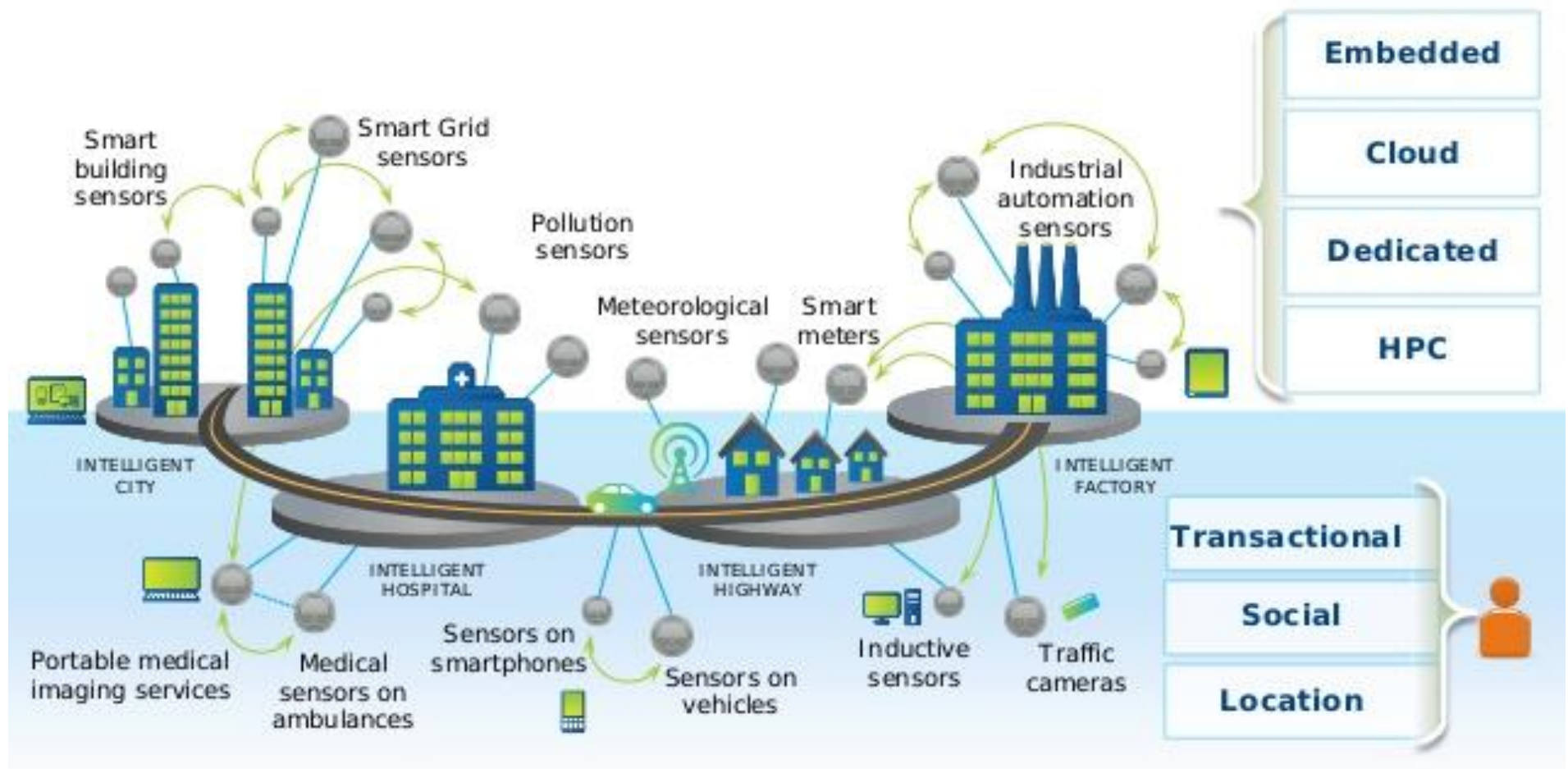
A Smart City is a city well performing in 6 characteristics, built on the 'smart' combination of endowments and activities of self-decise, independent and aware citizens.



SMART CITIES FOR SENSING

Smart cities and sensing data (Internet of Things - IoT)

Smart City Sensor Model



SMART CITY SCENARIO

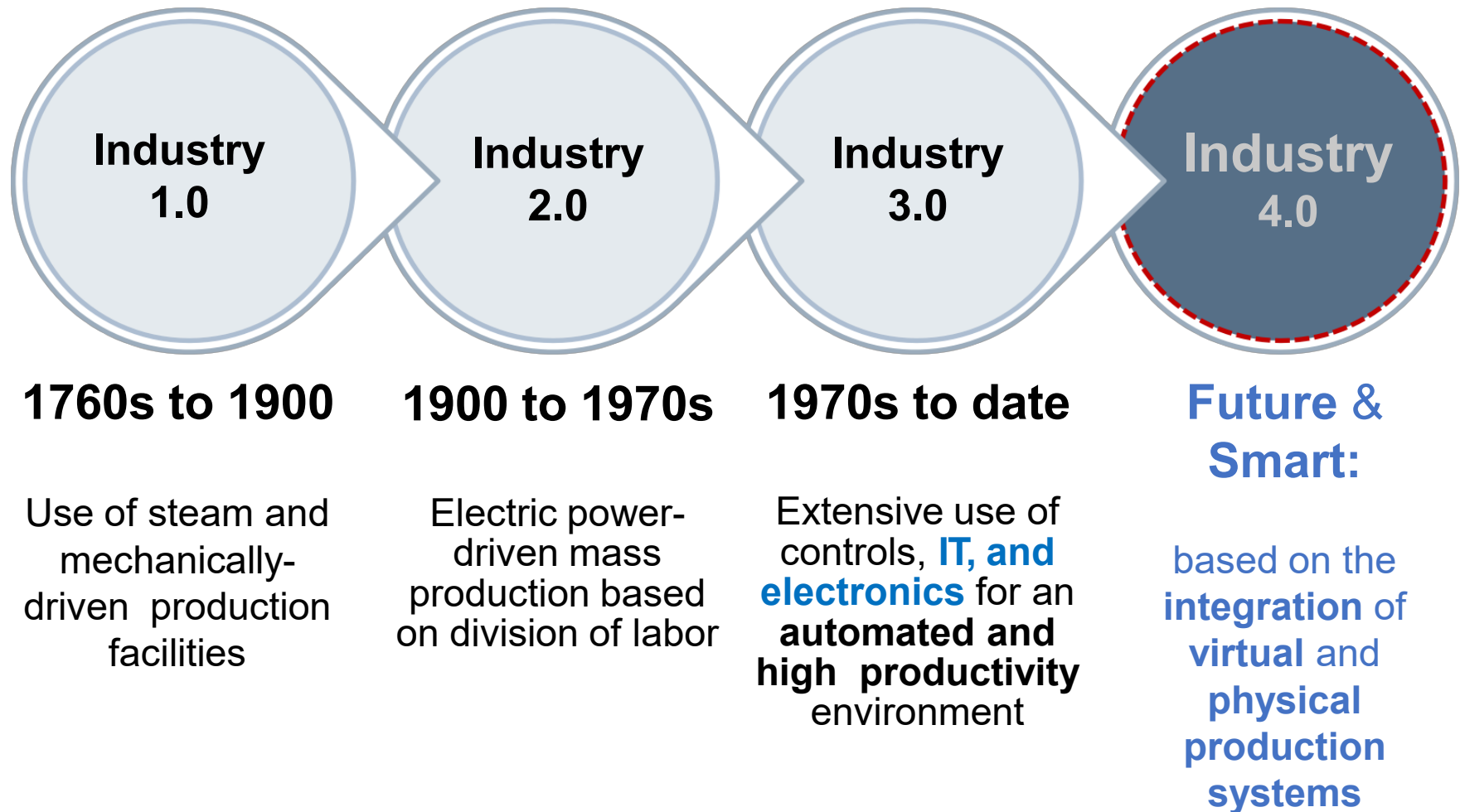
In a **smart city**, we may consider and pay **attention** to some **specific behaviors that produce a big data system in interaction with other ones** (**within the complexity stemming from global interaction**)

- **Group of replicated resources** and interacting components
- **Co-creation of new contents** such as videos, pictures, etc.
- **Collection of big data**
- **Harvesting of open data**
- **Management of resources and people information**
- **Public services**
- **Specific workflow for communities**
- **Logistic in the cities**

We can also **focus on some localities** to work with and test and experience **a smaller-size isolated system**

INDUSTRY 4.0

Industry 4.0 is in the trends of the **industrial revolutions** focusing more on the big data implied and managed



INDUSTRY 4.0 – EUROPE 2013

Industry 4.0 was a spreading trend toward an evolution of traditional **industrial processes** and it became a reality

Smart industry or “INDUSTRIE 4.0” refers to the **technological evolution from embedded systems to cyber-physical systems**

Industry 4.0 (I4.0) has multiple meanings

- connects / merges **production with ICT**
- merges **customer data with machine data**
- goes **M2M**: Machines communicate with Machines
- **components and machines** autonomously manage **production in a flexible, efficient, and resource-saving manner**

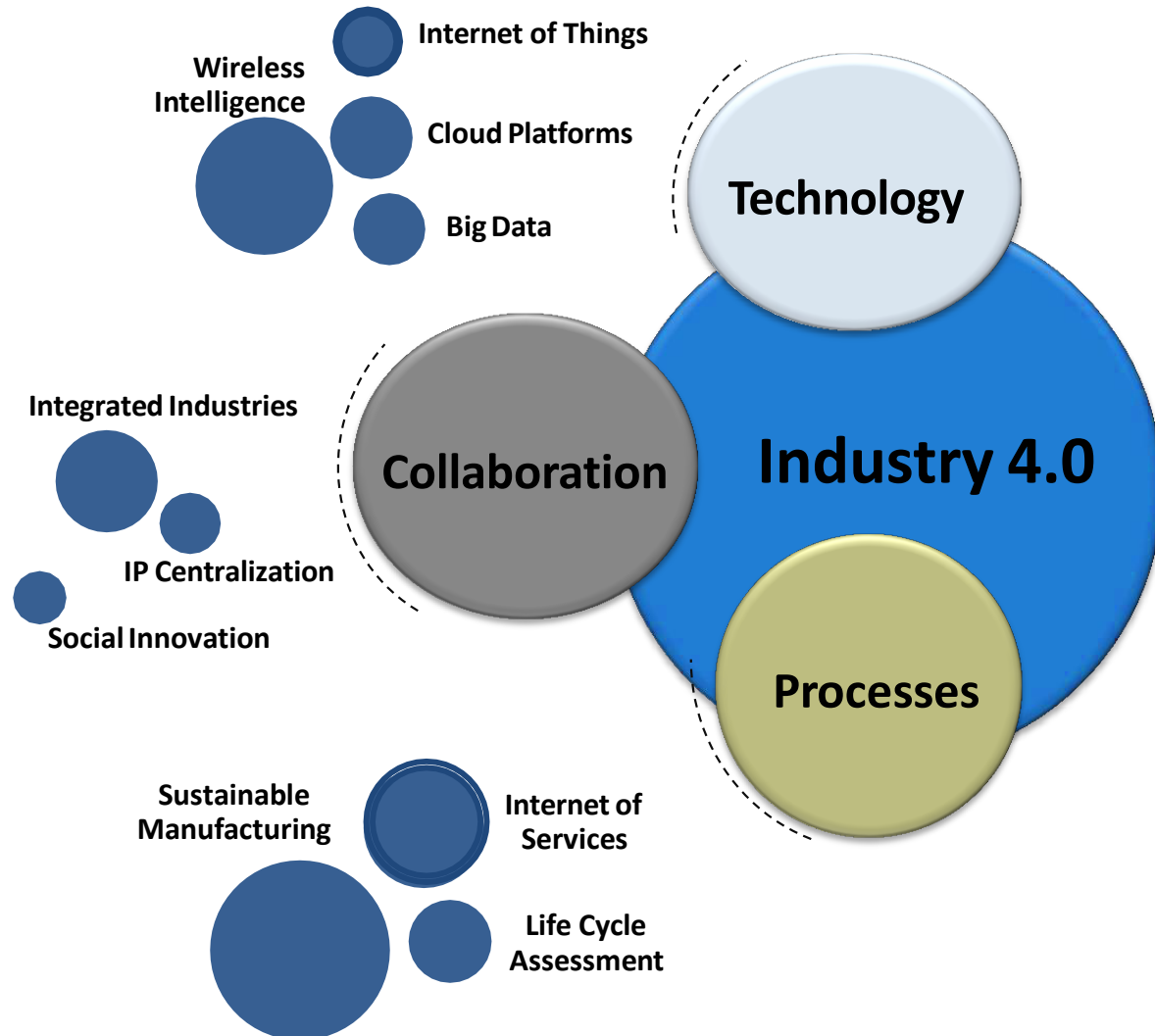


INDUSTRY 4.0 ENVIRONMENT

Industry 4.0 is in the sense of **product innovation in manufacturing**

as an effort
in three areas

- **Technology**
- **Collaboration**
- **Processes**



INDUSTRY 5.0 – EU COMMISSION DEFINITION

Industry 5.0 encompasses manufacturing to extend the scope of focus on society with specific more societal goals (**2021**)

- **Human-centric**
- **Sustainable**
- **Resilience**



INDUSTRY 5.0 – EU COMMISSION



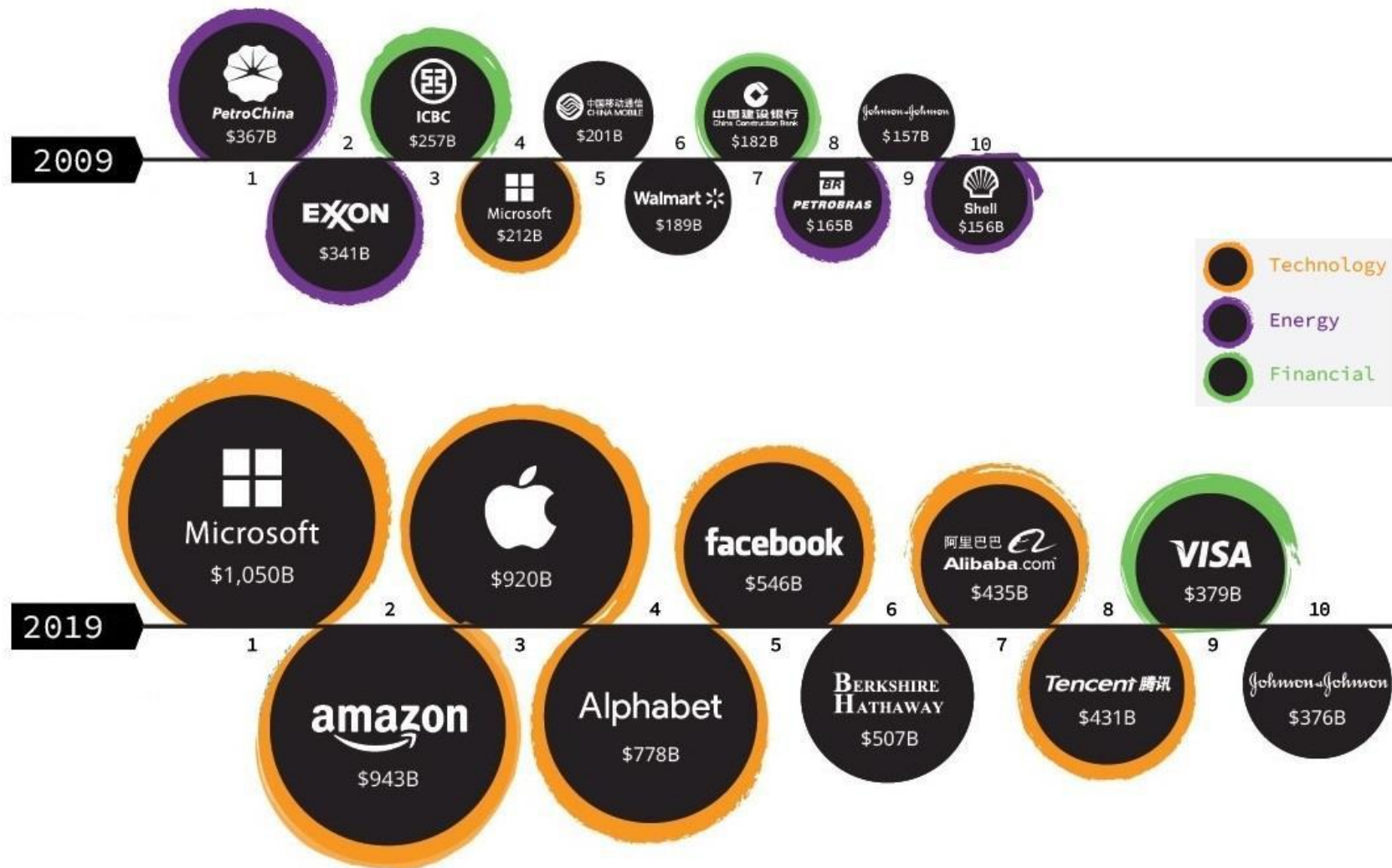
Industry 5.0 encompasses manufacturing to extend with specific more societal goals

- **Human-centric** - human needs and interests at the heart of the **production process**. *Rather than asking what we can do with new technology, we ask what the technology can do for us.* Rather than asking the industry worker to adapt skills to the needs of rapidly evolving technology, we want to use technology to adapt the production process to the needs of the worker
- **Sustainable** - develop circular processes that **recycle natural resources, reduce waste and environmental impact**. *Sustainability means reducing energy consumption and greenhouse emissions*
- **Resilience** - develop a higher degree of **robustness in industrial production**, against disruptions and support critical infrastructure in times of crisis, and overcome the fragility of our current approach to globalized production, by developing sufficiently **resilient strategic value chains, adaptable production capacity and flexible business processes**, *especially where value chains serve basic human needs, such as healthcare or security*

COMPANIES AND DATA IN A DECADE

Largest market Companies ---- trends

Change in main core



2026 COPILOT

As of early 2026, the largest 10 companies in the world by market capitalization are:

Rank	Company	Market Cap (USD)	Sources
1	NVIDIA (... , semicond)	\$4.17T–\$4.56T	[alpha-sense.com] , [financecharts.com] , [fool.com]
2	Alphabet (Google)	~\$3.89T–\$4.00T	[financecharts.com] , [fool.com] , [indianexpress.com]
3	Apple	~\$3.51T–\$3.85T	[financecharts.com] , [fool.com] , [indianexpress.com]
4	Microsoft	~\$3.33T–\$3.55T	[financecharts.com] , [fool.com] , [indianexpress.com]
5	Amazon	~\$2.37T–\$2.63T	[financecharts.com] , [fool.com] , [indianexpress.com]
6	TSMC (Taiwan semicond)	~\$1.38T–\$1.78T	[alpha-sense.com] , [financecharts.com]
7	Broadcom (... , semicond)	~\$1.50T–\$1.67T	[alpha-sense.com] , [financecharts.com]
8	Meta Platforms	~\$1.48T–\$1.63T	[alpha-sense.com] , [financecharts.com] , [fool.com]
9	Saudi Aramco (oil gas)	~\$1.52T–\$1.61T	[indianexpress.com] , [companiesm...ketcap.com]
10	Tesla	~\$1.34T–\$1.46T	[financecharts.com] , [fool.com]
			Starting

2025 COPILOT

As of early 2025, the largest companies in the world by market capitalization are:

- 1.Apple** - \$3.496 trillion
- 2.Microsoft** - \$3.092 trillion
- 3.NVIDIA** - \$2.874 trillion
- 4.Amazon** - \$2.531 trillion
- 5.Alphabet (Google)** - \$2.511 trillion
- 6.Saudi Aramco** - \$1.793 trillion (oil/gas)
- 7.Meta Platforms (formerly Facebook)** - \$1.742 trillion
- 8.Tesla** - \$1.317 trillion
- 9.Broadcom** - \$1.034 trillion
- 10.Berkshire Hathaway** - \$1.009 trillion (finance/conglomerate)

2026 COPILOT

2026 - comparison of IT vs non IT giants

Largest Companies by Market Capitalization

IT		Non-IT	
 nVIDIA	\$4.2T	 Saudi Aramco	\$1.5T
Alphabet	\$3.9T	Berkshire Hathaway	\$1.1T
 Apple	\$3.6T	Walmart 	\$1.0T
 Microsoft	\$3.4T	JPMORGAN CHASE &	\$0.8T
amazon 	\$2,4T		

INTERESTING TARGET SYSTEMS FOR ICCBD

There are many **interesting applications in interesting systems**

We are **interested in connected and distributed systems**

in-the-small

Limited size

A set of locations closely interconnected with several application running but limited in distance and high-bandwidth connections and limited communication times

Local installations

in-the-large

Multicloud Big Data

A very large set of interconnected locations with a non-countable number of interacting applications depending on each others

In general, global world-wide systems

COURSE CORE

The complexity of applications in-the-large asks for
ready-to-use off-the-shelf solutions

The answer toward a better usage is “**Middleware**”

We can give a **first definition**

Middleware is a **set of tools** and **components** already available for the best **system performance**, mainly under the **user required perspective (data and processing)**

A **middleware** can make **available ready-to-use applications** if a user needs new functions with no user intervention

A **middleware** can also **simplify the development of new applications** if the functions are not already available

A **middleware** can also follow **life cycle to adapt the system** to **new requirements and trends**

PRIVATE & COMPANY SUPPORT

A **middleware to support the needs and requirements** of either a **private or public organization with specific goals to provide services to internal users**

- **Has several user machines and applications** (traditional PCs, mobile & small group resources, ...)
- **Works on company server in local data center** (typically servers and their connected resources)
- **Must access to remote resources** (either on other companies or on global Cloud)

Examples of needed support services/functions:

- **Transparent services**: replication/group synch, load balancing, naming, accountability,
- **Non-Transparent Project management and support tools**: service monitoring, decision systems, ...
mainly company-based and company-negotiated
- **Management of service delivery & used resources** (computing, storage, network, ...): if it is a **technology company**, both via **CLI** and visual **UI**

CLOUD PROVIDERS

The **second large case** is a **middleware to support the needs and requirements of a (general-purpose) data center typically available in Cloud**

- **Has several IT internal resources** (large quantities of servers in groups, large data servers and storage, more special purpose IT resources, ...)
- **Offers services to several customer organizations** (typically bare services, and more articulated ones)
- **Has to honor accepted contracts** (not only locally, but also coordinating with provider in need)

CLOUD PROVIDERS

The main goal is to have one **global offer**
middleware to provide services to any external customer with very differentiated QoS

- only **locally provided**, but also capable of **coordinating with other providers** in need

Examples of needed support services/functions:

- **Management & monitoring of physical infrastructure & of support functions** to enable sharing of resources
- **Advanced physical resource management to grant** agreed quality levels, isolation (security & performance)
- **Customer Relationships Management (CRM)**, for all services to interact with customer needs, ...

The **internal resource setting** must give **services** with the **required and agreed upon QoS to the external customers**

CLOUD AS AN EVOLUTION

A necessary and unavoidable step ahead
Cloud Architectures and solutions

Possibility of **off-the-shelf solutions** organized around and with **Web-accessible resources** in **remote data centers** via **services** the customer can ask for

- **ready-to-use** services
- **easy** services
- **pay-per-use** services
- **transparent** (or *non-transparent*) services
- flexible, **extensible & elastic** services
- **reliable** services
- **secure** services

REQUIREMENT FOR SERVICES

In distributed systems, while services must be correctly provided, a critical goal is the **Quality of Service (QoS)**, in the sense of provisioning with some parameters and respecting some requirements

The **QoS** has many different meanings, because it is a very **general quality indicator**

It can stress **response time, correctness, availability, confidence, security, user satisfaction, ...**

QoS goals (conflicting?) in the **Old** and the **New World**

- **Old world:** typically, main goals **reliability** and **enforced consistency, ACID**
- **New world:** **scalability and availability** matters **most of all**

Focus on **extremely fast response times: Amazon** estimates that **each millisecond** of delay has a measurable impact on sales!

BEHIND THE SCENES: SUPPORT FOR...

To **provide QoS** distributed systems have to support some coverage of **properties** and **functions for granting the value to information and data (metadata)**

- **Meaning**: data must be wherever interpreted and understood
- **Replication**: usage of multiple copies of resources
- **Grouping**: keeping together different copies and behavior
- **Simplified delivery**: new tools and technologies to fasten development & deployment of complex applications
- **Automated management**: infrastructures taking care of management burden with minimal human intervention
- **Batch data processing**: storage/processing of massive amounts of data, such as for Google Web indexing
- **Streaming data**: dealing with information series coming from a set of grouped info, such as a video, sensors, etc.

THE ICCBD CLASS TOPICS

The course aims at **elaborating on the knowledge of distributed systems for the whole life cycle operation**, for the aspects related the execution

- **Operations** in the **entire life cycle**
- **System management**
- **Quality of Service (QoS)**
- **Observation and monitoring of all resources**
- **Replication** and **Partitioning of localities and resources**
- **Dynamic variations of resources** during the **life cycle**
- **Recovery** and **tuning**
- **Extreme disaster recovery of resources**
- **Sustainability** and **resilience**

DISTRIBUTED SYSTEMS AND APPLICATIONS

Middleware to support Distributed Systems

Where a suitable infrastructure (a middleware) handles and manages all system resources

Some interesting Middleware lines

- **Object** middleware (**CORBA**, COM, .NET, ...)
- **Message exchange** middleware (**MOM**)
- **Overlay Networks, Novel File systems, NoSQL support**
- **Cloud systems & middleware** (OpenStack, ...)
- **Middleware for Big Data:** *storing* - new data repositories, *processing & streaming* (NoSQL, Hadoop, SPARK, Cassandra, MongoDB, ...)

Middleware as environments to support services

Some tools are common to all different kinds of middleware

CLASS INTERESTS

Topics oriented toward the execution environment

- **All the aspects** are selected in the sense of their contribution toward a **better execution**
- **General topics** are explored for their **presence and support for the execution part of the life cycle**, always **dominant in time**

Less interest paid to

- Design phases, coding, preparation and static analysis

Individual starting experience

- Capacity of **reading technical papers**
- Skill to **support going depth into a topic**
- **Writing & Presentation** on technical topics
- **Design** a small project and solution sketch

INPUT REQUIREMENTS

Skills on **operations in different environments**
(previous lab presence is recommended)

Skills on **most significant models for distributed systems**
concurrency, processing, storage, ...

LATERAL SKILLS

Capacity of **implementing** and **controlling** real projects

Capacity of **exploring in an independent way**

Skills in **project engineering**

Skills in **English** ...

GOALS

Design of a service/application architecture
Execution and performance of the project

Analysis Capacities

- Understanding of **Principles** and **support environments** for general-purpose services and special-purpose ones
- Understanding of **Projects** and **Solutions** at different levels: conceptual, architectural, at protocol level, algorithmic one, by using different technologies & components

Synthesis Capacities (see site)

- Speech based on some read **paper**, chosen & elaborated
- Design of a ***chosen case study***
- Presentation of a **written report** as a 'to-be-published' article

Orienteering Capacity

CLASS ASSESSMENT

The **final grading** stems from an **oral exam**

to ascertain the **knowledge** and **orientation** about the entire discipline, ranging on all topics, starting with the basics, going through the practical portions of middleware, and also with a possible follow-up on a chosen topic

Also, it is possible deepening on a specific agreed practical topic with some added (microprojects)

You can also choose the project activity (for 3 credits), recommended for the Distributed System Computer Engineering path

Assignment of a project on a specific subject assigned for anyone and done individually

A Report and a Presentation for the exam

PROJECT ACTIVITIES

Projects can deal with any topics of the class for 3 credits

And **Big Data** and **Cloud continuum** areas of **industrial interests**

- **Smart Cities**
- **Smart Tourism**
- **Industry 5.0**
- **Critical Infrastructures**
- **Cloud continuum Observability**
- **QoS-aware Observability for critical systems**
- **Cloud continuum Network management**
- **Cloud Infrastructure for IoT provisioning**
- **QoS-aware microservice composition in Service Meshes**
- **Serverless Storage**
- **Hardware acceleration in Service Mesh**
- ...

GRADING - WORKFLOW

The final score is via the oral exam (*almaesami* for enrollment)

For **exam booking**, first step is the enrollment on the list

*Scheduled days in **almaesami** and **oral exams** for the class on:*

- First exam (Wednesday, 18th June 2026)
- Second exam (Thursday, 3rd July 2025)
- Third exam (Thursday, 24th July 2025)

For **project activities** look at these dates, give in the project, then we define a date for discussion

- The same *scheduled days in **almaesami*** for giving in the two-part project (**report & implemented project**)
- Then you are expected to make a **presentation** of it

HANDS-ON SEMINARS

We have classes:

Wednesday 12.30 – 15.30

classroom III

Friday 9 - 12

classroom 4.1

Plan of hands-on experience about some novel directions in relevant technologies not within class hours

Define slots: Friday - at 13.30 /14 ??? classroom 5.1 ???

Possible a few seminars to introduce company technology perspective

Companies can give a picture of what is their experience, and which technical roles are significant for them and within their organization

Importance of

Possibility of studying abroad / work experience

Serious language skills (apart from technical)

SOME MATERIALS AND ITEMS

Class Slides available:

- on the web site of the class

Some basic books

G. Coulouris, J. Dollimore, T. Kindberg, "***Distributed Systems: Concepts and Design***", Addison-Wesley, (fifth edition) 2012.

([find pdf on line](#))

M. Kleppmann:, "**Designing Data-Intensive Applications**", O'Reilly, 2017. ([find pdf on line](#))

A.S. Tanenbaum, M.v.Steen "***Distributed Systems: Principles and Paradigms***", Prentice-Hall, second edition 2006/2013.

B. Forouzan, F. Mosharraf: "***Computer Networks, a top-down approach***", McGraw-Hill, 2011.

M.L. Liu, "***Distributed Computing***", Addison-Wesley, 2003.

SOME (CLASSIC) REFERENCE BOOKS

- D.L. Galli, "***Distributed Operating Systems: Concepts and Practice***", Prentice-Hall, 2000.
- L. Peterson, B. Davie, "***Computer Networks, A Systems Approach***", Second edition, Morgan Kaufmann, 2000.
- V.K. Garg, "***Elements of Distributed Computing***", Wiley, 2002.
- J.F. Kurose, K.W. Ross, "***Computer Networking: a Top-Down Approach Featuring the Internet***", McGraw-Hill, 2001).
- J. Siegel, "***CORBA 3: Fundamentals and Programming***", (second edition), OMG Press, Wiley, 2000.
- F. Halsall, "***Multimedia Communications***", Addison-Wesley, 2001.
- T. Erl *et al.*, "***Cloud computing : concepts, technology, & architecture***", Prentice Hall, 2013.

SOME (CLASSIC) REFERENCE BOOKS

- B. Wilder, “***Cloud architecture patterns***”, Beijing, 2013.
- A. T. Velte *et al.*, “***Cloud computing: a practical approach***”, McGraw-Hill, 2010.
- J. Rhoton, “***Cloud computing explained***”, Recursive Press, 2009.
- T. Fifield *et al.*, “***Openstack operations guide: set up and manage your OpenStack cloud***”, O'Reilly, 2014.
- S. Holla, “***Orchestrating Docker***”, Packt Publishing, 2015.
- O. Hane, “***Build your own PaaS with Docker***”, Packt Publishing, 2015.
- T.D. Nadeau and K. Gray, “***SDN: software defined networks***”, O'Reilly, 2013.
- L. Carlson, “***Programming for PaaS***”, O'Reilly, 2013.
- T. White, “***Hadoop: the definitive guide***”, O'Reilly, 2012.

SOME (CLASSIC) REFERENCE BOOKS

- E. Sammer, “***Hadoop operations***”, O'Reilly, 2012.
- K. Rankin, “***DevOps troubleshooting***”, Addison-Wesley, 2013.
- D. Sui *et al.*, “***Crowdsourcing geographic knowledge***”, Springer, 2013.
- Z. Yan *et al.*, “***Semantics in mobile sensing***”, Morgan & Claypool, 2014.
- R. Copeland, “***MongoDB applied design patterns***”, O'Reilly, 2013.
- J. Carpenter, “**Cassandra: The Definitive Guide**”, 3rd Edition, O'Reilly, 2020
- Silvano Gai, “**Building a Future-Proof Cloud Infrastructure: A Unified Architecture for Network, Security, and Storage Services**”, Addison-Wesley, 2020
- Brendan Gregg, “**Systems Performance**”, 2nd Edition, Pearson, 2020

MANY SOURCES – INTERNET APART

Please refer to articles on different topics in journals published by two professional organization:

- **ACM (Association for Computing Machinery)** e
 - **IEEE (Institute of Electrical and Electronic Engineering)**
- Groups www.computer.org www.comsoc.org

General magazine:

- *IEEE Computer, ACM Communications*
- [IEEE Internet Computing](#) e [IEEE Communications](#)

Depth into journals very specific and helpful

- **ACM Computing Surveys** (ACM CS), **ACM Transactions on...**
- [IEEE Transactions on](#) (IEEE Trans..., ACM Trans...)
- IETF Request For Comments **RFC**